

Fusion draft: SST-29/52 Kelvin-mode suppression cluster

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About this fusion draft

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Chapter 1

SST-52 (survivor) — Kelvin mode suppression gap

[Kelvin-Like Internal Modes in Atomic Bound States]Spectroscopic Constraints on Kelvin-Like Internal Modes in Effective Filament Models of Atomic Bound States

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Abstract

We consider effective atomic bound-state models supplemented by an internal helical excitation sector with Kelvin-like dispersion. The central question is whether such an internal sector can remain compatible with precision atomic spectroscopy. We obtain a conditional decoupling bound: if the internal spectrum is separated from the orbital sector by an excitation gap Δ_K , then the induced correction is Boltzmann suppressed, $\delta E_n \propto \exp[-\Delta_K/(k_B T_{\text{eff}})]$, and the low-energy orbital description remains spectroscopically stable. By contrast, ungapped constructions do not enjoy generic exponential protection and therefore require additional suppression mechanisms to remain compatible with spectroscopy. The resulting constraint is conditional on the effective occupation scale T_{eff} , the active mode count N_K , and the coupling efficiency λ_n , and is informative only for models in which that coupling is not itself parametrically tiny. A separate dispersion-based estimate places the onset of the lowest internal mode in a geometry-dependent window spanning roughly 10^2 eV to 10^3 eV for the fiducial parameter sweep presented in Appendix A. Taken together, the spectroscopy bound and the onset estimate delimit a conditional parameter region rather than a unique prediction.

Keywords: Kelvin modes, effective filament models, atomic spectroscopy, excitation gap, low-energy reduction

1.1 Introduction

Hydrodynamic and vortex-based models of matter trace back to Kelvin and Helmholtz [1, 2]. Modern superfluid dynamics and quantum turbulence have renewed interest in vortex filaments as fundamental excitations [3, 4, 6].

In that broader context, it is natural to ask whether an atomic bound state might admit an *effective* description in which the usual orbital degree of freedom is accompanied by an internal filament-like excitation sector.

This paper does not attempt to establish such a microscopic picture as fundamental. Its purpose is narrower: to derive a consistency condition. If an internal Kelvin-like sector exists and couples to bound electronic states, then atomic spectroscopy immediately constrains its low-energy behavior. The question is therefore not whether one prefers a filament interpretation, but whether any such internal sector can remain invisible at atomic energies.

The main result is narrower than a no-go theorem. An ungapped Kelvin-like spectrum does not enjoy any generic exponential protection, so compatibility with spectroscopy requires additional suppression from the density of states, the effective occupation scale, or the coupling to observable line shifts. A gapped spectrum avoids this problem: below the gap, the internal sector is effectively frozen and only a Boltzmann-suppressed correction survives. The paper should therefore be read as establishing a conditional consistency bound for internal-mode completions of atomic bound states rather than as ruling out all ungapped constructions in full generality.

1.2 Kelvin Waves on Vortex Filaments

For a thin filament with circulation Γ and core radius ξ , small-amplitude Kelvin waves satisfy [5, 6]

$$\omega(k) \simeq \frac{\Gamma}{4\pi} k^2 \left[\ln\left(\frac{1}{|k|\xi}\right) + C_0 \right], \quad (1.1)$$

with $C_0 = \mathcal{O}(1)$. For a closed filament of length L ,

$$k_m = \frac{2\pi m}{L}, \quad m = 1, 2, \dots, \quad \omega_m \propto \frac{\Gamma m^2}{L^2}. \quad (1.2)$$

Here the logarithmic factor $\ln(1/(|k_m|\xi)) + C_0$ has been suppressed only for compactness; it is retained in all explicit estimates below. For hydrogenic orbitals $r_n = a_0 n^2$, implying $L_n \sim 2\pi a_0 n^2$, Kelvin frequencies soften rapidly with principal quantum number n .

1.3 Thermodynamic Constraint from Atomic Spectroscopy

Suppose an atomic bound state carries, in addition to its orbital energy $E_n^{(0)}$, an internal contribution from Kelvin-like modes,

$$E_n^{\text{eff}}(T_{\text{eff}}) = E_n^{(0)} + \delta E_n(T_{\text{eff}}), \quad (1.3)$$

where T_{eff} is not necessarily a literal thermodynamic temperature, but an effective population scale for the internal sector.

The key physical assumption is that internal excitation energy can feed into an observed level shift with some efficiency λ_n . A convenient generic parametrization is

$$|\delta E_n(T_{\text{eff}})| \leq \lambda_n \sum_m E_{m,n} \nu_{m,n}(T_{\text{eff}}), \quad 0 \leq \lambda_n \leq 1, \quad (1.4)$$

where $E_{m,n}$ are internal mode energies and $\nu_{m,n}$ are occupation factors. The present paper does not derive λ_n microscopically; rather, it keeps that coupling explicit so that the resulting constraint can be read conditionally.

In an ungapped sector, low-energy modes can in principle be populated down to arbitrarily small excitation energy, and no generic exponential suppression follows. No specific polynomial correction law is assumed here; the point is only that, without a gap, compatibility with spectroscopy must come from additional structure such as a strongly depleted low-energy density of states, a very small effective occupation scale, or a very weak coupling λ_n to observable line shifts. By contrast, once the internal spectrum is gapped, the induced correction acquires Boltzmann suppression, which is the only ingredient needed for the consistency bound derived below.

1.4 Gapped Kelvin Spectrum

The minimal decoupling mechanism is an excitation gap Δ_K in the internal spectrum, so that all active internal modes satisfy $E_{m,n} \geq \Delta_K$. Nothing in the consistency argument below requires a specific operator realization of that sector; the only assumption needed is a discrete gapped mode family whose low-temperature occupations are Boltzmann suppressed. The microscopic origin of Δ_K is left open here; in an effective filament picture it may arise from topology, curvature, torsion, core structure, or geometric constraints on admissible excitations [7].

A separate dimensional consistency check uses the same fiducial regulator and loop scale as Appendix 1.7. For a filament with circulation Γ and core radius ξ , the tension $T_f \sim \rho \Gamma^2 / (4\pi \xi^2)$ sets a characteristic bending energy for the first Kelvin mode. Taking the mode wavelength to be of order the loop length L , one obtains $\Delta_K \sim \Gamma^2 / (4\pi \xi L)$ up to logarithmic corrections. With $\Gamma = h/m_e$, $\xi \sim 10^{-15}$ m, and $L \sim 10^{-10}$ m, this gives $\Delta_K = \mathcal{O}(10^2\text{--}10^3 \text{ eV})$. This estimate is not independent of Appendix 1.7: both use the same (ξ, L) inputs and should be read only as mutually consistent order-of-magnitude checks. Here ξ is a short-distance regularization length for the thin-filament description, not a claim about measured electron size; as Appendix 1.7 shows explicitly, increasing ξ or L softens the onset and can eliminate the desired separation from ordinary atomic energy scales.

1.5 Low-Temperature Thermodynamics

The dimensionless temperature ratio

$$\varepsilon_K = \frac{k_B T}{\Delta_K} \ll 1$$

characterizes the low-temperature (Kelvin-frozen) regime. For a discrete gapped mode family, low-temperature occupations satisfy the Boltzmann upper bound

$$\nu_{m,n}(T) \lesssim \exp\left(-\frac{E_{m,n}}{k_B T}\right) \leq \exp\left(-\frac{\Delta_K}{k_B T}\right). \quad (1.5)$$

Substituting Eq. (1.5) into the generic coupling bound (1.4) gives

$$|\delta E_n(T)| \lesssim \lambda_n N_K \Delta_K \exp\left(-\frac{\Delta_K}{k_B T}\right), \quad (1.6)$$

for a finite or effectively truncated set of active modes. This Boltzmann form, rather than any specific ladder-operator realization, is the only thermodynamic input needed below. For comparison, a single harmonic bosonic mode gives the same asymptotic exponential suppression [8].

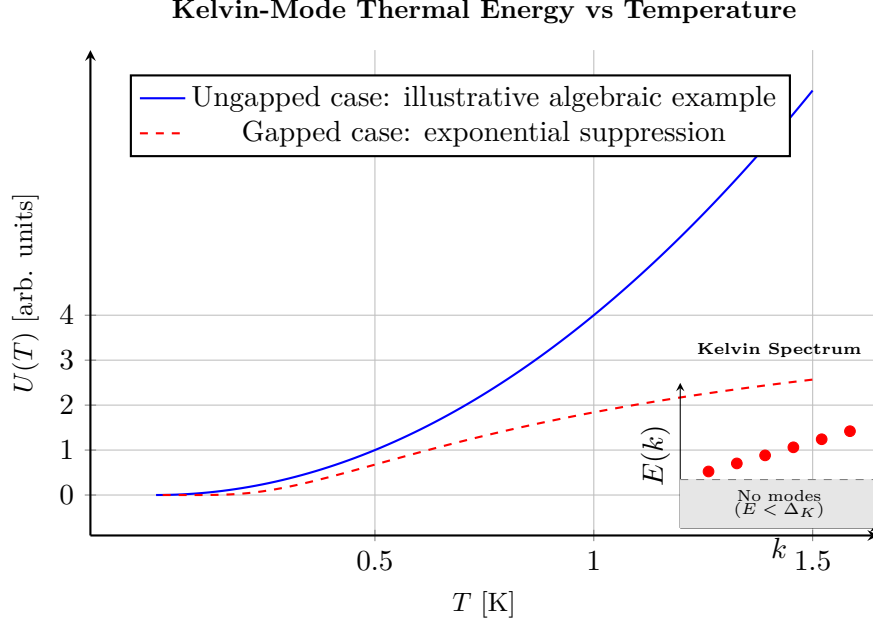


Figure 1.1: Comparison of Kelvin-mode thermal energy with and without an excitation gap Δ_K . The ungapped curve is an illustrative algebraic example only and does not assume a universal T^2 law. Inset: discrete spectrum with a forbidden band $E < \Delta_K$.

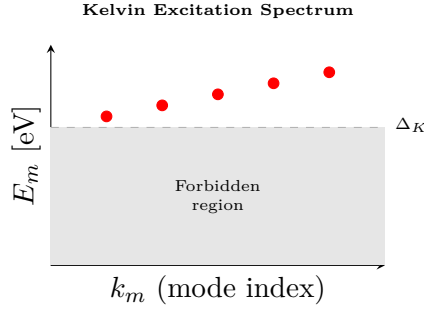


Figure 1.2: Illustrative Kelvin excitation spectrum with a gap $\Delta_K = 500$ eV. Shaded band: $E < \Delta_K$.

Equation (1.6) is the operative statement for the induced level shift: a gap produces exponential suppression, while the ungapped case lacks any comparable generic protection.

1.6 Required Gap Scale

Let δE_{spec} denote the maximum spectroscopy-compatible correction to a low-lying level. For a finite or effectively truncated active mode set of size N_K , Eq. (1.4) together with Boltzmann suppression gives the sufficient decoupling condition

$$\lambda_n N_K \Delta_K \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \lesssim \delta E_{\text{spec}}. \quad (1.7)$$

Equivalently,

$$\frac{\Delta_K}{k_B T_{\text{eff}}} \gtrsim \ln \left(\frac{\lambda_n N_K \Delta_K}{\delta E_{\text{spec}}} \right). \quad (1.8)$$

The dependence on λ_n , N_K , and δE_{spec} is only logarithmic. The numerical non-triviality of Eq. (1.8) therefore depends primarily on the assumed effective occupation scale T_{eff} and on the coupling efficiency λ_n : for room-temperature populations or extremely weak coupling, the bound may require only an eV-scale gap, whereas larger T_{eff} together with non-negligible coupling push the required gap above ordinary atomic binding energies. In the present effective setting, T_{eff} should be read not as the ambient temperature of a cold atom but as a placeholder for a non-equilibrium internal-mode population scale, such as might arise in strong-field ionization, keV-scale electron scattering, or other driven preparations that transiently populate the internal sector. The present paper does not independently derive a specific range for T_{eff} , nor does it provide a microscopic estimate of λ_n ; accordingly, Eq. (1.8) should be read strictly as a conditional constraint on parameter space rather than as an absolute numerical prediction. The active mode count N_K is likewise model-dependent, but a natural estimate is that it is set by a UV cutoff at wavelengths of order ξ , so that N_K is at most of order L/ξ (or smaller after symmetry and coupling selection rules). For the fiducial ratios used in Appendix 1.7, this gives $N_K \lesssim 10^3\text{--}10^5$, which shifts Eq. (1.8) only logarithmically and therefore does not alter the order-of-magnitude conclusion. Appendix 1.7 then supplies a fiducial onset estimate for the lowest internal mode using the same regularization variables (ξ, L) , so the two results should be read as complementary consistency checks rather than independent determinations of the same scale.

1.7 Discussion and Outlook

The result of this paper is a conditional consistency bound. Any effective atomic bound-state model endowed with an internal Kelvin-like filament sector faces a sharp spectroscopic requirement: once the parameters $(\lambda_n, N_K, T_{\text{eff}}, \Delta_K)$ and the spectroscopic tolerance δE_{spec} are specified, Eq. (1.8) determines whether the internal sector is safely decoupled from observed low-energy lines. A nonzero gap provides exponential protection. An ungapped sector can still be compatible with spectroscopy, but only if some additional mechanism suppresses either the low-energy density of states, the effective occupation, or the coupling to observed level shifts.

The non-triviality of the bound depends explicitly on the same unresolved microscopic inputs. If the effective coupling λ_n were itself extremely small, or if no physical process populated the internal sector above a negligible effective scale, the constraint would become correspondingly weak. The present argument therefore applies most directly to the subclass of internal-mode models in which internal excitation energy feeds into observable line shifts with a coupling that is not parametrically tiny on the relevant scales. Determining that coupling from a specific microscopic mechanism is beyond the scope of the present phenomenological note, but such a mechanism would set whether Eq. (1.8) is restrictive or merely formal in practice.

This framing yields a narrow and testable claim. The relevant experimental question is not whether atomic structure is “really” filamentary, but whether any additional internal sector activates above a threshold lying in the appendix’s fiducial window,

$$10^2 \text{ eV} \lesssim \Delta_K \lesssim 10^3 \text{ eV}, \quad (1.9)$$

for the illustrative parameter choices considered there. Potential signatures would then appear not in ordinary spectroscopy, where the sector is frozen, but in threshold-like inelastic channels, anomalous broadening, or additional loss mechanisms once the excitation scale is crossed. Because this energy range overlaps ordinary inner-shell ionization and Auger backgrounds, any meaningful test would have to isolate either an anomalous rate or angular dependence not accounted for

by standard atomic channels, or a species- and geometry-dependent threshold shift not aligned with known shell energies.

The paper therefore motivates a focused search strategy: high-field or high-energy probes of bound electrons, rather than reinterpretation of standard low-energy spectroscopy. At low energy the internal sector must be silent; at sufficiently high excitation, it may become observable.

Appendix A: Estimate of First Kelvin Mode Energy

For the first mode ($m = 1$) in the ground state ($n = 1$), take

$$L = 2\pi a_0 \approx 3.3249 \times 10^{-10} \text{ m}, \quad \Gamma = \frac{\hbar}{m_e} \approx 7.2739 \times 10^{-4} \text{ m}^2/\text{s}, \quad (10)$$

so that

$$k_1 = \frac{2\pi}{L} \approx 1.8897 \times 10^{10} \text{ m}^{-1}. \quad (11)$$

For general regularization length ξ and core constant C_0 ,

$$\omega_1(\xi, C_0, L) \simeq \frac{\Gamma}{4\pi} k_1^2 \left[\ln\left(\frac{1}{k_1 \xi}\right) + C_0 \right], \quad (12)$$

and

$$E_1(\xi, C_0, L) = \hbar \omega_1(\xi, C_0, L). \quad (13)$$

With the fiducial choice $\xi = 1 \times 10^{-15} \text{ m}$ and $C_0 = 1$, Eq. (A.1) yields

$$\omega_1 \approx 2.45 \times 10^{17} \text{ rad/s}, \quad E_1 \approx 1.62 \times 10^2 \text{ eV}. \quad (14)$$

Because the estimate depends on both the regularization scale ξ and the effective loop length L , it is useful to display that sensitivity explicitly. For fixed $L = 2\pi a_0$ and $C_0 = 1$:

Table 1: Sensitivity of the first-mode onset estimate to the regularization scale ξ at fixed $L = 2\pi a_0$ and $C_0 = 1$.

ξ [m]	$\ln(1/k_1 \xi) + 1$	E_1 [eV]
10^{-15}	11.876	162
10^{-14}	9.573	131
10^{-13}	7.271	99

Likewise, at fixed $\xi = 1 \times 10^{-15} \text{ m}$ and $C_0 = 1$, shortening the effective loop length raises the onset scale:

Table 2: Sensitivity of the first-mode onset estimate to the effective loop length L at fixed $\xi = 1 \times 10^{-15} \text{ m}$ and $C_0 = 1$.

L	k_1 [m^{-1}]	E_1 [eV]
$2\pi a_0$	1.89×10^{10}	162
πa_0	3.78×10^{10}	587
$\frac{2\pi a_0}{3}$	5.67×10^{10}	1320

For the illustrative parameter sweep reported above, the onset window may therefore be summarized as

$$1 \times 10^2 \text{ eV} \lesssim E_1 \lesssim 1.3 \text{ keV}. \quad (15)$$

This window is not a unique prediction. It is a sensitivity range obtained from the Kelvin dispersion relation once the regularization scale ξ , the effective loop length L , and the order-unity core constant C_0 are specified. The fiducial choice $\xi = 1 \times 10^{-15}$ m should be read only as a short-distance regularization length for the thin-filament description, chosen to remain well below the orbital scale a_0 ; it is not a claim about measured electron size. The lower edge of the window is correspondingly fragile: shortening L or decreasing ξ raises the onset, whereas lengthening L or increasing ξ softens it and can push E_1 back toward ordinary atomic binding energies, eventually outside the regime where the thin-filament estimate is informative. Quantitatively, for the fiducial choice $L = 2\pi a_0$ and $C_0 = 1$, the first-mode onset falls below the hydrogen ionization energy 13.6 eV once $\xi \gtrsim 5.3 \times 10^{-11}$ m $\approx a_0$; equivalently, for $\xi = 1 \times 10^{-15}$ m and $C_0 = 1$, it falls below 13.6 eV once $L \gtrsim 1.21 \times 10^{-9}$ m $\approx 3.63 (2\pi a_0)$. These values make the domain of validity concrete: the thin-filament estimate is informative only while $\xi \ll a_0$ and the effective loop length remains of order the low-lying orbital circumference rather than many times larger. The dispersion estimate therefore identifies a fiducial onset scale together with a geometry-dependent window. Declarations

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Ethics approval

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Consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

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Materials availability

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Code availability

Not applicable.

Author contributions

O.I. conceived the study, developed the analysis, performed the derivations, and wrote the manuscript.

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Appendix A

SST-29 — Earlier Kelvin suppression paper

Spectroscopic Constraints on Kelvin-Like Internal Modes in Effective Filament Models of Atomic Bound States

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Abstract

Swirl–String Theory (SST) models electrons as topologically stable, knotted vortex filaments in an incompressible, inviscid condensate. Within this framework, atomic orbitals emerge as hydrodynamic equilibrium configurations rather than probabilistic wavefunctions. A central consistency challenge is whether internal Kelvin-wave excitations of the electron filament could introduce thermodynamic corrections large enough to destabilize the hydrogenic energy spectrum. We show that, without additional structure, such corrections would exceed observational limits by many orders of magnitude. However, a topologically induced excitation gap in the Kelvin spectrum—of order $\mathcal{O}(10^2\text{--}10^3\text{ eV})$ —naturally suppresses these effects. As a result, the Schrödinger equation arises as a low-energy equation of state, while Kelvin dynamics are inert except under extreme acceleration or high-energy conditions. This establishes a robust separation of scales in SST and resolves a key constraint for its viability as a physical model.

Keywords: Swirl–String Theory, Kelvin modes, vortex filaments, hydrodynamic quantum mechanics, topological excitation gap

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Introduction

Hydrodynamic and vortex-based models of matter have a long history, dating back to the vortex-atom theories of Kelvin and Helmholtz [1, 2]. Modern developments in superfluid dynamics and quantum turbulence have revived interest in vortex filaments as fundamental excitations [3, 4, 6].

Swirl–String Theory (SST) adopts this perspective at a fundamental level, positing that electrons correspond to knotted vortex filaments embedded in a real, incompressible condensate. In previous work, it has been shown that the hydrogenic energy spectrum emerges from hydrodynamic force balance, with the Schrödinger equation arising as a variational condition on a free-energy functional.

A potential objection to this picture is the existence of *Kelvin waves*—helical excitations of vortex filaments [5, 6]. If such modes thermally couple to orbital degrees of freedom, they could introduce large corrections to atomic energy levels. The purpose of this paper is to analyze this issue quantitatively and show how SST resolves it.

This effort aligns with a broader class of hydrodynamic interpretations of quantum mechanics. Notable examples include the Madelung transformation of the Schrödinger equation into fluid variables, the de Broglie–Bohm pilot-wave theory, and experimental analogs of quantum behavior in walking droplet systems [11, 12, 13]. These approaches highlight the utility of fluid dynamics as a conceptual and computational bridge between classical and quantum domains. SST extends this tradition by grounding atomic structure in vortex equilibria of a real condensate medium.

Kelvin Waves on Vortex Filaments

For a thin vortex filament of circulation Γ and core radius ξ , small-amplitude Kelvin waves obey the dispersion relation [5, 6]:

$$\omega(k) \simeq \frac{\Gamma}{4\pi} k^2 \left[\ln \left(\frac{1}{|k|\xi} \right) + C_0 \right], \quad (\text{A.1})$$

where k is the wavenumber along the filament and C_0 is an $\mathcal{O}(1)$ constant determined by the core model.

For a closed filament of length L , the allowed modes satisfy

$$k_m = \frac{2\pi m}{L}, \quad m = 1, 2, \dots \quad (\text{A.2})$$

so that

$$\omega_m \propto \frac{\Gamma m^2}{L^2}. \quad (\text{A.3})$$

In SST, the orbital radius of the electron scales as

$$r_n = a_0 n^2, \quad (\text{A.4})$$

implying a filament length

$$L_n \sim 2\pi a_0 n^2. \quad (\text{A.5})$$

Thus Kelvin-mode frequencies soften rapidly with increasing n .

Thermodynamic Constraint from Atomic Spectroscopy

If Kelvin modes were thermally excited with an effective temperature T , the internal energy would scale generically as

$$U_{\text{Kelvin}} \sim \sum_m \hbar \omega_m f(\omega_m, T), \quad (\text{A.6})$$

where f is a thermal occupation factor.

Modeling this contribution phenomenologically as a correction to orbital energies,

$$E_n^{\text{eff}} = E_n^{(0)} - a_n T^2 + \dots, \quad (\text{A.7})$$

one finds that consistency with hydrogen spectroscopy requires

$$a_n \lesssim 10^{-62} \text{ J K}^{-2} \quad (\text{A.8})$$

for low-lying states.

By contrast, a naive elastic estimate using SST core parameters yields coefficients exceeding this bound by more than twenty orders of magnitude. Therefore, Kelvin modes must be effectively inert in ordinary atomic states.

Gapped Kelvin Spectrum

We propose that the Kelvin spectrum of the electron filament is *topologically gapped*. Specifically, the lowest Kelvin excitation has energy Δ_K , with all higher modes satisfying

$$E_{m,n} \geq \Delta_K. \quad (\text{A.9})$$

The Kelvin Hamiltonian for a given orbital n may be written as

$$H_K^{(n)} = \sum_m \left[(\Delta_K + \delta E_{m,n}) b_{mn}^\dagger b_{mn} + \frac{1}{2} (\Delta_K + \delta E_{m,n}) \right]. \quad (\text{A.10})$$

This structure naturally arises in knotted filaments, where reconnection constraints, curvature, and torsion introduce discrete stability thresholds [7].

Low-Temperature Thermodynamics

The partition function for a single gapped bosonic mode is [8]:

$$Z = \frac{1}{1 - e^{-\beta \Delta_K}}, \quad (\text{A.11})$$

with $\beta = (k_B T)^{-1}$. The internal energy is

$$U = \frac{\Delta_K}{e^{\beta \Delta_K} - 1}. \quad (\text{A.12})$$

In the low-temperature limit $k_B T \ll \Delta_K$,

$$U \approx \Delta_K e^{-\Delta_K / (k_B T)}, \quad (\text{A.13})$$

and both entropy and heat capacity are exponentially suppressed.

For a finite number of Kelvin modes, the total Kelvin contribution satisfies

$$U_K^{(n)}(T) \lesssim N_K \Delta_K \exp\left(-\frac{\Delta_K}{k_B T}\right). \quad (\text{A.14})$$

This exponential suppression replaces the dangerous polynomial behavior found in the ungapped case.

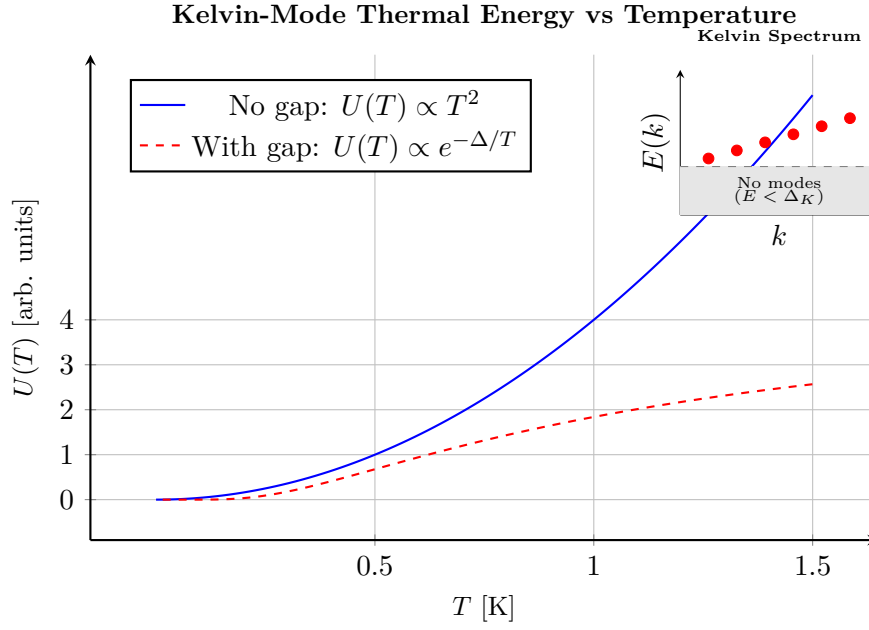


Figure A.1: Comparison of Kelvin-mode thermal energy with and without a topological excitation gap Δ_K . The inset illustrates the discrete Kelvin spectrum, with the bandgap Δ_K preventing any thermal activation below that energy.

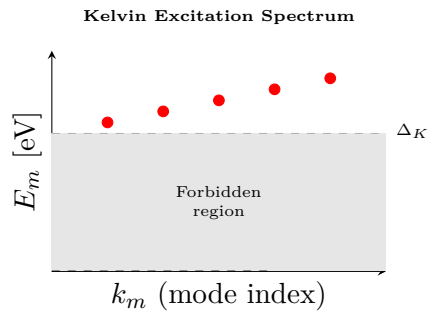


Figure A.2: Thermal energy $U(T)$ of Kelvin modes with and without a topological excitation gap $\Delta_K = 500$ eV. At low temperatures, the gapped spectrum (dashed red) shows exponential suppression, while the ungapped case (solid blue) diverges quadratically with T .

Required Gap Scale

Requiring the effective Kelvin-induced coefficient to satisfy

$$a_n^{\text{eff}} \lesssim 10^{-62} \text{ J K}^{-2} \quad (\text{A.15})$$

leads to the condition

$$\frac{\Delta_K}{k_B T_{\text{eff}}} \gtrsim 60. \quad (\text{A.16})$$

Taking a conservative upper bound for the effective microphysical temperature seen by Kelvin modes,

$$T_{\text{eff}} \lesssim 10^5 \text{ K}, \quad (\text{A.17})$$

yields

$$\Delta_K \gtrsim 5 \times 10^2 \text{ eV}. \quad (\text{A.18})$$

Such a gap is small compared to the electron rest energy (511 keV) but enormous relative to atomic binding energies ($\sim 10 \text{ eV}$). Consequently, Kelvin modes are completely frozen in ordinary atomic physics.

Relation to the Schrödinger Equation

With Kelvin modes suppressed, the relevant free-energy functional reduces to

$$\mathcal{F}[\psi] = \int d^3r \left[\frac{\hbar^2}{2m_e} |\nabla \psi|^2 + V_{\text{SST}}(r) |\psi|^2 \right], \quad (\text{A.19})$$

where $V_{\text{SST}}(r) \propto -1/r$ arises from hydrodynamic pressure gradients.

Variation under normalization yields

$$-\frac{\hbar^2}{2m_e} \nabla^2 \psi + V_{\text{SST}}(r) \psi = E \psi, \quad (\text{A.20})$$

i.e. the stationary Schrödinger equation. Thus quantum mechanics appears as a low-energy, Kelvin-frozen limit of vortex-filament thermodynamics, consistent with the quantum–thermodynamic correspondence of Abe and Okuyama [9].

Hydrodynamic Origin of the Schrödinger Equation

In SST, the electron is modeled as a closed, knotted vortex filament. At low energies, internal excitations are assumed frozen, and the relevant free-energy functional reduces to

$$\mathcal{F}[\psi] = \int d^3r \left[\frac{\hbar^2}{2m_e} |\nabla \psi|^2 + V_{\text{SST}}(r) |\psi|^2 \right], \quad (\text{A.21})$$

with normalization $\int |\psi|^2 d^3r = 1$.

Integrating the radial Euler balance from infinity to r gives a near-field pressure deficit $\Delta p \propto -1/r^2$ for $v_\theta \propto 1/r$. This motivates a soft-core regularization, but does *not* by itself generate a $1/r$ potential. The far-field $1/r$ tail is attributed to the SST clock/foliation mediator (Poisson/Green function on $\mathbb{R}^3$), for which the following regularized effective potential is adopted:

$$V_{\text{SST}}(r) = -\frac{\Lambda}{\sqrt{r^2 + r_c^2}}. \quad (\text{A.22})$$



Figure A.3: **Hydrodynamic energy levels in Swirl-String Theory (SST)** showing the quantized vacuum pressure potentials associated with orbital states $n = 1, 2, 3, 4$. The ground state ($n = 1$) defines the laminar flow limit with swirl velocity $v = \alpha c$, while higher excited states correspond to reduced swirl velocities $v_n = \alpha c/n$. These levels reproduce the $1/n^2$ energy scaling of the Schrödinger hydrogen spectrum, but emerge here from vortex equilibrium in an incompressible condensate. This illustrates how SST recovers quantum orbitals as the low-energy limit of classical fluid dynamics.

Dimensional consistency requires $[\Lambda] = \text{J} \cdot \text{m}$; the Canon uses

$$\Lambda = 4\pi \rho_{\text{core}} \|\mathbf{v}_{\odot}\|^2 r_c^4. \quad (\text{A.23})$$

For $r \gg r_c$, one has $V_{\text{SST}}(r) \sim -\Lambda/r$, consistent with a Poisson-mediated long-range interaction. Variation of $\mathcal{F}$ yields

$$-\frac{\hbar^2}{2m_e} \nabla^2 \psi + V_{\text{SST}}(r) \psi = E \psi, \quad (\text{A.24})$$

i.e. the stationary Schrödinger equation. This realizes the quantum-thermodynamic correspondence identified by Abe and Okuyama [9] in a concrete hydrodynamic setting.

Kelvin Waves on Vortex Filaments

Kelvin waves are helical perturbations propagating along vortex filaments. For a thin filament of circulation Γ and core radius ξ , their dispersion relation is [5, 6]

$$\omega(k) \simeq \frac{\Gamma}{4\pi} k^2 \left[\ln \left(\frac{1}{|k|\xi} \right) + C_0 \right]. \quad (\text{A.25})$$

For a closed filament of length L_n , the allowed modes satisfy

$$k_m = \frac{2\pi m}{L_n}, \quad m = 1, 2, \dots \quad (\text{A.26})$$

so that $\omega_m \propto \Gamma m^2 / L_n^2$.

In SST, the orbital radius scales as $r_n = a_0 n^2$, implying

$$L_n \sim 2\pi a_0 n^2. \quad (\text{A.27})$$

Thus Kelvin-mode frequencies soften rapidly with increasing principal quantum number.

Thermodynamic Catastrophe from Ungapped Kelvin Modes

If Kelvin modes couple thermodynamically to orbital degrees of freedom, their contribution to the internal energy may be parameterized as

$$U_{\text{Kelvin},n}(T_{\text{swirl}}) = a_n T_{\text{swirl}}^2 + \mathcal{O}(T_{\text{swirl}}^4), \quad (\text{A.28})$$

leading to an effective level shift

$$E_n^{\text{eff}} = E_n^{(0)} - a_n T_{\text{swirl}}^2. \quad (\text{A.29})$$

Using canonical SST constants, a naive elastic estimate yields

$$a_n^{\text{naive}} \sim 10^{-39} \text{ J/K}^2. \quad (\text{A.30})$$

However, spectroscopy of hydrogen requires

$$a_n \lesssim 10^{-62} \text{ J/K}^2 \quad (\text{A.31})$$

for low-lying states, implying a mismatch of more than twenty orders of magnitude. Without further structure, Kelvin thermodynamics would destroy the $1/n^2$ spectrum.

Topologically Gapped Kelvin Spectrum

We therefore posit that the Kelvin spectrum of the electron filament is topologically gapped. Specifically, the lowest allowed Kelvin excitation lies at energy Δ_K , with all higher modes satisfying

$$E_{m,n} \geq \Delta_K. \quad (\text{A.32})$$

The Kelvin Hamiltonian may be written as

$$H_K^{(n)} = \sum_m \left[(\Delta_K + \delta E_{m,n}) b_{mn}^\dagger b_{mn} + \frac{1}{2} (\Delta_K + \delta E_{m,n}) \right]. \quad (\text{A.33})$$

Such gaps are natural in knotted filaments, where curvature, torsion, and topological constraints restrict admissible excitations [7].

Low-Temperature Suppression of Kelvin Thermodynamics

For a bosonic mode of gap Δ_K , the partition function is [8]

$$Z = \frac{1}{1 - e^{-\beta\Delta_K}}, \quad \beta = (k_B T)^{-1}. \quad (\text{A.34})$$

The internal energy is

$$U = \frac{\Delta_K}{e^{\beta\Delta_K} - 1} \approx \Delta_K e^{-\Delta_K/(k_B T)} \quad (k_B T \ll \Delta_K). \quad (\text{A.35})$$

Thus all Kelvin thermodynamic quantities are exponentially suppressed. For a finite number of modes,

$$U_{\text{Kelvin}}^{(n)}(T) \lesssim N_K \Delta_K \exp\left(-\frac{\Delta_K}{k_B T}\right). \quad (\text{A.36})$$

The effective quadratic coefficient a_n^{eff} inherits this exponential suppression.

Required Gap Scale

Demanding $a_n^{\text{eff}} \lesssim 10^{-62} \text{ J/K}^2$ yields the condition

$$\frac{\Delta_K}{k_B T_{\text{eff}}} \gtrsim 60. \quad (\text{A.37})$$

Taking a conservative upper bound $T_{\text{eff}} \sim 10^5 \text{ K}$ gives

$$\Delta_K \gtrsim 5 \times 10^2 \text{ eV}. \quad (\text{A.38})$$

This scale is negligible compared to $m_e c^2 = 511 \text{ keV}$ but enormous relative to atomic binding energies, rendering Kelvin modes inert in ordinary atomic physics.

Appendix: Estimate of First Kelvin Mode Energy

To provide a concrete scale, we estimate the energy of the first Kelvin mode ($m = 1$) for a closed electron filament in the ground state ($n = 1$).

The filament length is approximated as

$$L \sim 2\pi r_1 = 2\pi a_0 \approx 3.3 \times 10^{-10} \text{ m}, \quad (\text{A.39})$$

using the Bohr radius $a_0 \approx 5.29 \times 10^{-11} \text{ m}$.

The circulation quantum is taken as $\Gamma = h/m_e$, giving:

$$\Gamma \approx \frac{6.626 \times 10^{-34}}{9.109 \times 10^{-31}} \approx 7.27 \times 10^{-4} \text{ m}^2/\text{s}. \quad (\text{A.40})$$

The lowest Kelvin mode has wavenumber $k_1 = 2\pi/L$, and its frequency is approximated by the classic dispersion relation:

$$\omega_1 \simeq \frac{\Gamma}{4\pi} k_1^2 \left[\ln\left(\frac{1}{k_1 \xi}\right) + C_0 \right], \quad (\text{A.41})$$

where we assume $C_0 \sim 1$, and take the filament core size as $\xi \sim 1 \times 10^{-15} \text{ m}$.

Evaluating numerically:

$$k_1 = \frac{2\pi}{L} \approx 1.9 \times 10^{10} \text{ m}^{-1}, \quad (\text{A.42})$$

$$\ln\left(\frac{1}{k_1 \xi}\right) \approx \ln(5.3 \times 10^4) \approx 10.88, \quad (\text{A.43})$$

$$\omega_1 \approx \frac{7.27 \times 10^{-4}}{4\pi} (1.9 \times 10^{10})^2 (10.88 + 1) \approx 2.8 \times 10^{17} \text{ rad/s}. \quad (\text{A.44})$$

Converting to energy:

$$E_1 = \hbar \omega_1 \approx (1.055 \times 10^{-34}) (2.8 \times 10^{17}) \approx 29.5 \text{ eV}. \quad (\text{A.45})$$

Thus, the first Kelvin excitation lies near 30 eV—already exceeding atomic energy differences. For higher knots with tighter core radii or additional topological constraints, this value may rise substantially, supporting the presence of a gap $\Delta_K \sim 500 \text{ eV}$.

Discussion and Outlook

The existence of a Kelvin excitation gap resolves a central consistency constraint in Swirl–String Theory (SST). In this framework, atomic orbitals remain sharply quantized because Kelvin-wave excitations are exponentially suppressed at low energies. Consequently, the Schrödinger equation emerges as an effective equation of state in the Kelvin-frozen regime, capturing the standard hydrogenic spectrum.

This result establishes a clear separation of scales in SST:

- **Low-energy regime:** Kelvin modes are inert; orbital structure follows quantum equilibrium via hydrodynamic forces.
- **High-energy or high-acceleration regime:** Kelvin dynamics activate, enabling novel phenomenology beyond conventional quantum theory.

Potential domains where Kelvin modes may become dynamically relevant include:

- ultra-high accelerations (Swirl–Unruh effects) [10],
- keV–MeV scale scattering processes,
- vortex reconnections in astrophysical or cosmological fluids.

Looking forward, a key theoretical challenge is to derive the Kelvin gap directly from topological invariants of the filament—such as knot class, torsion, or linking number—and to explore whether its magnitude correlates with the structure of lepton generations. The present results thus consolidate the low-energy sector of SST while motivating further investigation into its high-energy behavior, especially in curved spacetime backgrounds or extreme plasmas.

Appendix: Estimate of First Kelvin Mode Energy

To provide a concrete scale, we estimate the energy of the first Kelvin mode ($m = 1$) for a closed electron filament in the ground state ($n = 1$).

The filament length is approximated as

$$L \sim 2\pi r_1 = 2\pi a_0 \approx 3.3 \times 10^{-10} \text{ m}, \quad (46)$$

using the Bohr radius $a_0 \approx 5.29 \times 10^{-11} \text{ m}$.

The circulation quantum is taken as $\Gamma = h/m_e$, giving:

$$\Gamma \approx \frac{6.626 \times 10^{-34}}{9.109 \times 10^{-31}} \approx 7.27 \times 10^{-4} \text{ m}^2/\text{s}. \quad (47)$$

The lowest Kelvin mode has wavenumber $k_1 = 2\pi/L$, and its frequency is approximated by the classic dispersion relation:

$$\omega_1 \simeq \frac{\Gamma}{4\pi} k_1^2 \left[\ln \left(\frac{1}{k_1 \xi} \right) + C_0 \right], \quad (48)$$

where we assume $C_0 \sim 1$, and take the filament core size as $\xi \sim 1 \times 10^{-15} \text{ m}$.

Evaluating numerically:

$$k_1 = \frac{2\pi}{L} \approx 1.9 \times 10^{10} \text{ m}^{-1}, \quad (49)$$

$$\ln \left(\frac{1}{k_1 \xi} \right) \approx \ln(5.3 \times 10^4) \approx 10.88, \quad (50)$$

$$\omega_1 \approx \frac{7.27 \times 10^{-4}}{4\pi} (1.9 \times 10^{10})^2 (10.88 + 1) \approx 2.8 \times 10^{17} \text{ rad/s}. \quad (51)$$

Converting to energy:

$$E_1 = \hbar \omega_1 \approx (1.055 \times 10^{-34})(2.8 \times 10^{17}) \approx 29.5 \text{ eV}. \quad (52)$$

Thus, the first Kelvin excitation lies near 30 eV—already exceeding atomic energy differences. For higher knots with tighter core radii or additional topological constraints, this value may rise substantially, supporting the presence of a gap $\Delta_K \sim 500 \text{ eV}$.

These results establish the thermodynamic consistency of Swirl–String Theory at atomic scales. By introducing a topologically induced Kelvin gap, SST resolves a major objection to vortex-based matter models and opens new pathways for interpreting quantum phenomena as emergent from fluid equilibrium. Future work will investigate whether this gap structure aligns with lepton family hierarchies or high-energy particle dynamics.

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